Dr. Roberto Mozara

Energy Systems Modeler & Scientific Software Developer

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Profile

Physicist (Dr. rer. nat.) with experience in energy system simulation (pandapower, PyPSA), battery storage modelling, and scientific software development. Strong Python and HPC background. Seeking a technical role in energy systems or software engineering.

Professional Experience

Energy Systems Modeler / Founder, *BESS Advisory Services* — 10/2024 – 08/2025, Hamburg
- Developed Python models for PV–battery operation in DA/ID markets
- Simulated grids with pandapower & PyPSA (load flow, congestion)
- Automated power-flow runs and data processing
- Modeled battery charge/discharge performance

Postdoctoral Researcher (Computational Condensed Matter Physics), — 10/2022 – 03/2024, Hamburg
I. Institute for Theoretical Physics, University of Hamburg
- Simulated quantum materials using HPC resources
- Developed scientific software in Python, C++, and Fortran
- Combined DFT approaches with advanced correlation techniques

Teacher of Physics and Mathematics, *Ulrich-von-Hutten Gymnasium* — 07/2019 – 07/2022, Berlin
- Taught physics and mathematics at upper and middle-school levels
- Designed and conducted exams, exercises, and learning modules
- Explained complex concepts clearly and didactically

Research Associate / PhD Researcher, — 06/2013 – 07/2018, Hamburg
I. Institute for Theoretical Physics, University of Hamburg
- Researched correlated electron systems and quantum materials
- Developed and used scientific simulation software (TRIQS, VASP)
- Collaborated with national and international research groups
- Published peer-reviewed papers and presented results

Visiting Researcher, *Institute of Applied Physics, University of Hamburg* — 02/2013 – 05/2013, Hamburg
- Applied algebraic methods to spin dynamics
- Worked with the Magneto-Theory research group

Education

PhD in Physics (magna cum laude), *University of Hamburg* — 2013 – 2021, Hamburg
Thesis: *Realistic Multi-Orbital Systems and Electronic Correlations*

Diploma in Physics (equivalent to MSc), *University of Hamburg & DESY* — 2004 – 2012, Hamburg
Focus: Mathematical physics, quantum field theory
Grade: very good

Exchange Semester, *University of Buenos Aires, Argentina (2007)* — 2007, Buenos Aires

Technical Skills

Programming Languages
Python, C++, Fortran, HTML/CSS, JavaScript (intro), Bash

Energy Systems & Simulation
pandapower, PyPSA, Pyomo (intro), battery storage optimization, charge/discharge modelling

Scientific Computing
DFT, DMFT, TRIQS, VASP, HPC/cluster computing, parallelization, numerical methods

Data & Development Tools
pandas, numpy, scipy, matplotlib, Git, Linux/Unix, Jupyter, LaTeX

Languages

German — native • **English** — fluent • **Spanish** — basic

Projects

PV–Battery Optimization Model, *Python · Pyomo · pandas · NumPy*
- Implemented an operational model for battery charge/discharge scheduling in DA/ID markets
- Designed Python modules for data handling, time-series prep, and performance evaluation
- Analyzed market participation strategies and quantified potential revenue improvements

Power Grid Simulation & Analysis, *pandapower · PyPSA · Python*
- Built modular workflows for AC load flow and congestion analysis in distribution networks
- Developed Python routines for automated preprocessing, simulation runs, and visualization
- Investigated bottlenecks, voltage stability, and line loading under varying operating conditions

Scientific Computing Workflow for Quantum Materials, *DFT · TRIQS · HPC · Python*
- Performed large-scale simulations of correlated electron systems using HPC clusters
- Integrated DFT results with many-body methods (DMFT) for improved physical accuracy
- Automated calculation pipelines and post-processing for reproducible research workflows

Publications (selected)

Inelastic Tunneling into Multipolaronic Bound States in Single-Layer MoS_2
Physical Review X 15, 031030 (2025)

Atomically thin oxide layer on the elemental superconductor Ta(001) surface
Physical Review Materials 3, 094801 (2019)